SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 02

Giving AI Eyes on the Evidence



The Tool Use Pattern: instead of guessing from a prompt, the model calls real tools that read the artifacts, then reasons over what they return.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

From self-review to real evidence

1

What tools add

The model stops guessing and starts inspecting.

2

Before AI

Examiners opened each artifact by hand.

3

The problem

A model with no access invents the details.

4

The discipline

Ground every claim in a tool output.

5

The gap

Reasoning without access is still guessing.

6

Tool Use

The model requests a tool; the tool reads the data.

7

The case

A stolen SUV, a phone, and three tools.

8

Your lab

Call the tools, then let the agent call them.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

WHERE WE ARE

Lab 01 fixed the words. Lab 02 fixes the eyes.

Lab 01 · Reflection

The model reviewed its own draft and pulled overclaims back to the evidence. It improved how it wrote.

But it still only saw what was pasted into the prompt.

TODAY

Lab 02 · Tool Use

Now the model can call tools that open the real artifacts: the gallery, the metadata, the sale drafts.

It stops guessing about the evidence and starts reading it.

THE KEY IDEA

A tool is a function the model can call

ONE TOOL, THREE PARTS

A name
inspect_image_evidence

Typed inputs
file_name, case_description

An output
the record the model reads

WHY THIS MATTERS

The model never touches the phone data directly. It asks a tool, by name, with valid inputs. The tool reads the evidence and returns a result the model reasons over, so each answer rests on an observation you can check. It can still overstate it, so check its work.

IN THIS LAB

list_media_files finds photos inspect_image_evidence reads one photo inspect_listing_records reads sale drafts

THE MANUAL CRAFT

How an examiner inspects this phone

Open the gallery

Scroll the DCIM folder for photos near the theft time.

Read the metadata

Check each photo's capture time, GPS, and camera.

Check detections

Confirm the color, body type, and roof rack.

Open the drafts

See which photo is attached to a sale listing.

RELIABLE, BUT SLOW AND SERIAL

An examiner does this one artifact at a time, cross-checking by hand. It is accurate, but it does not keep up when there are thousands of photos and a model that could help, if only it could reach the data.

THE PROBLEM

A model with no access invents the details

ASK A PLAIN MODEL, WITH NO TOOLS

“IMG_3310.jpg, taken at 19:42, clearly shows the stolen SUV, and the listing was posted to Marketplace at 20:05.”

EVERY SPECIFIC IS FABRICATED

A filename that does not exist

The real candidate is IMG_2044.jpg. The model guessed.

A timestamp from nowhere

No artifact says 19:42. It was invented to sound precise.

A sale that never happened

No draft was ever published. “Posted” is pure hallucination.

THE DISCIPLINE

Do not ask the model. Ask the tool.

Ask the model what the photo shows

- The model guesses from the prompt.
- It invents filenames and times.
- The answer cannot be checked.

Ask a tool, then let the model reason

- The tool reads the real record.
- The model reasons over the result.
- Every claim traces to an output.

Same model, same case. The difference is whether a real observation sits under each claim.

THE GAP

The model can reason, but it cannot see

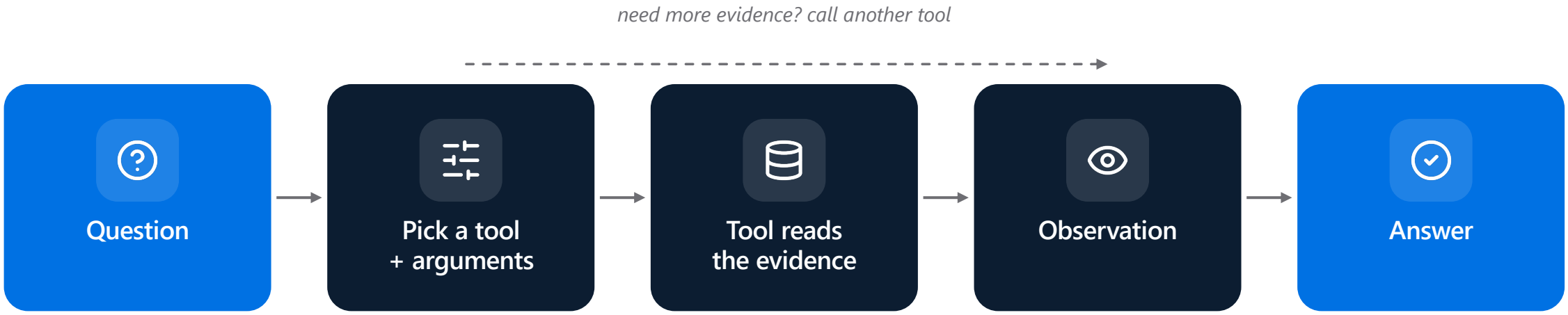


INTELLIGENCE IS NOT ACCESS

The model can weigh evidence as well as a junior analyst, but it has no hands and no eyes on the device. Reasoning over evidence it cannot read is just confident guessing. Tool Use is the bridge across this gap: it gives the model a safe, named way to reach the data.

THE ANSWER

The Tool Use Pattern



THE TOOLS SUPPLY THE DATA

The model picks the tool and its inputs. The facts come from the tool, but the model can still overstate them.

WHEN DOES IT STOP?

When it has enough observations it writes the report. If it answers before any tool has run, it is sent back to call one.

UNDER THE HOOD

How the model asks for a tool

THE MODEL WRITES A REQUEST

```
<tool_call>
{"name":
   "inspect_image_evidence",
 "arguments": {
   "file_name":
     "IMG_2044.jpg",
   "case_description":
     "black SUV with
      roof rack"}}
</tool_call>
```

The agent reads the request

It pulls the tool name and arguments out of the block.

It checks the arguments

Each value is coerced to the type the tool declared.

It runs the real function

The tool reads the evidence and returns a result.

The result goes back

The model reasons over it, or asks for another tool.

THE INVESTIGATION

Case TU-2026-004

THE DEVICE

PHONE

Samsung Galaxy S23, Android 14

ACQUIRED

Magnet AXIOM, logical + media

REPORTED STOLEN

black SUV with roof rack

RECOVERED

a phone used by the seller

THE QUESTION

Does the phone show that the stolen SUV was photographed and prepared for an online sale?

A photo
IMG_2044.jpg: a black SUV with a roof rack, captured 21:14 UTC.

A draft
A listing titled “black SUV for sale” with that photo attached, 21:31 UTC.

A boundary
The draft was never published. Preparation is not a completed sale.

YOUR INSTRUMENTS

Three tools read the evidence

list_media_files

root, date_from

Finds candidate photos in a folder, on or after a date.

inspect_image_evidence

file_name, case_description

Combines one photo's metadata and vehicle detection, and scores the match.

inspect_listing_records

date_from

Returns saved sale drafts and which photo each one attaches.

Core sequence: **list_media_files** -> **inspect_image_evidence** -> **inspect_listing_records**

SEE THE TOOLS BUILD A CONCLUSION

Three calls, one careful answer

1	**list_media_files("DCIM/Camera", from 2026-01-02)** -> 3 candidate photos	Narrows review to photos taken on or after the theft date.
2	**inspect_image_evidence("IMG_2044.jpg", "black SUV with roof rack")** -> strong match, 21:14 UTC	Black, SUV, and roof rack all agree with the stolen vehicle.
3	**inspect_listing_records(from 2026-01-02)** -> "black SUV for sale", IMG_2044 attached, 21:31 UTC	Links the same photo to a sale draft created minutes later.

Defensible conclusion: confirmed evidence the SUV was photographed and prepared for sale. The draft was never published, so a completed sale is not confirmed.

SAY EXACTLY HOW SURE YOU ARE

Confirmed, likely, or unconfirmed

Confirmed

Direct tool outputs support each part of the claim.

EXAMPLE

The SUV was photographed and attached to a sale draft.

Likely

The outputs suggest it, but one link is missing or weak.

EXAMPLE

A second photo matches, but its roof rack was not detected.

Unconfirmed

The outputs do not adequately support the claim.

EXAMPLE

That the listing was ever posted, or the vehicle sold.

PRACTICAL

Your turn, in the browser

1	Open + Part 0 setup	Click the Colab badge. Steps 0.1 to 0.4: install, paste a free Groq key (hidden, not saved), write the case files, load the tools.
2	Part A: call the tools	A.1 builds the three tools, A.2 shows the card the AI reads, A.3 runs them yourself, A.4 has the AI write up your evidence.
3	Part B: ToolAgent	B.1 builds the agent with the same tools and question. B.2 runs it: read every Using Tool, Tool call, and Tool result line.
4	Your Evidence Report	Five fields: tool-call log, strongest time, strongest vehicle match, a confirmed, likely, or unconfirmed label, and the limits.
5	Compare the two workflows	Three questions in their own answer cell: what the agent missed, what you accept or reject, what is not proven.

Before every code cell, a short note tells you its Purpose, its Input, How it works, and its Output.

SectraTEK

A model is only as good as what it can check.

Next lab: ReAct, where the agent reasons one step, acts, observes, and repeats until the answer holds.

